Research Pulse · Component map (as designed)

Steps 02 to 12 run unattended every Monday. Step 01 can happen any time and applies from the next run. Solid arrows are the main flow, dashed is optional.

	TRIGGER	INPUT	CONTEXT	PROCESSING	AI	OUTPUT
KAVITA Consultant, the user		**01** · Describes a domain in plain English; Gemini drafts categories, keywords and a note. She edits and saves. No code.				**15** · Reads it in under ten minutes and decides: read in full, skip, or bring to a client. The judgment stays with her.
VERCEL CRON Scheduler	**02** · Vercel Cron, Monday 02:00 UTC (about 07:30 IST, give or take an hour). Secret-checked GET. A Run-now button calls the same route.					
NEON Config and paper log			**03** · Domains table: categories, keywords, significance note, weight, active flag. Read at the start of every run.			**16** · Stores every shortlisted paper with its signals, verdict, summary, her feedback clicks and the run's counts.
ARXIV Candidate papers			**04** · arXiv: query the last 7 days, abstracts + authors. The only source that generates candidate papers.			
HF, S2, GITHUB Ranking signals only			Ranking signals, not a source: Hugging Face Daily Papers (upvotes) · Semantic Scholar h-index, OpenAlex fallback · GitHub (repo stars).			
FETCHER Vercel function · fetch				**05** · Per domain: query the last 7 days, up to 150 candidates, dedupe by arXiv id. One failed domain never stops the rest.		
RANKER Vercel function · rank				**06** · Score on cheap signals: keyword fit, upvotes, code link, stars, author record. Keep the top 12 per domain, drop the rest.		
GEMINI FLASH Judge and summarise			**07** · Rubric file (significance.md), the domain's own note, and its 12 abstracts. A JSON schema fixes the output shape.		**08** · One Flash call per domain: score 0 to 10, paper type, a two-line "so what", evidence strength, and read / skim / flag.	
COMPOSER Vercel function · compose				**09** · Keep 2 to 3 picks per domain above a threshold, or say "nothing significant". Cap near 1,800 words. Add scanned → shortlisted → shown.	**10** · Flash writes a three-line "this week in research" headline across all domains from the chosen picks.	**11** · Brief rendered once: an HTML email plus an archive page, picks grouped by domain, every card linked to its source.
RESEND Email delivery				**12** · Resend sends the email. The archive page is saved first, so a failed send never loses the brief. One retry.	*email last*	**13** · Optional: alert email to the owner if a stage failed or a domain came back empty.
BRIEF APP Next.js on Vercel						**14** · Archive page with Useful / Not relevant / Save-for-client buttons, a searchable paper log and CSV export.
MAINTAINER Pipeline owner				**18** · Backtest, available any time: replay any past week through steps 05 to 09 and check that papers known to matter surface.		**17** · Run-health page: per-run funnel (fetched → shortlisted → picked), useful-pick rate, failed domains and next cron time.

Human action · Storage / context · Deterministic processing · AI (judges, summarises, drafts) · Final output / commit · Optional

Gemini Flash is the only AI in the system and is called in three places: the settings helper (01, shown in blue because she makes the final edit), per-domain judging (08) and the headline (10). Every other step is plain code, so a Gemini outage degrades the brief (ranked list, no summaries) instead of stopping it. The system never claims a paper is validated.

Not pictured (shipped after this map was drawn): Stage 5 upgrades — claim-check / deep read on the top pick per domain, OpenReview + OpenAlex context for the judge, and a GitHub Actions cron backstop — plus the client-export digest at /export, which turns "Save for client" picks into a downloadable/copyable brief.